

The General Variance Conjecture from the Isotropic Thin-Shell Theorem

Literature-implied answer note for arXiv:1209.4270

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Status. Literature-implied answer (full; agent-identified implication). Conjectures 1.4 and 1.5 of Alonso-Gutierrez–Bastero are true in all dimensions. The later supporting paper proves the isotropic theorem but does not state this anisotropic consequence.

Abstract

Alonso-Gutierrez and Bastero conjectured that every centered log-concave random vector X satisfies

$$\mathrm{Var}(|X|^2) \leq C \|\mathrm{Cov}(X)\|_{\mathrm{op}} \mathbb{E}|X|^2.$$

Klartag and Lehec recently proved the isotropic variance conjecture in every dimension. We show by Abel summation over the eigenvalues of $\mathrm{Cov}(X)$ that the isotropic theorem implies the displayed general inequality with exactly the same best dimension-free constant. Hence the source’s general variance and general thin-shell conjectures are completely settled. The implication is elementary but is not stated in the supporting paper found in the bounded literature search.

1 Source questions

The source is D. Alonso-Gutierrez and J. Bastero, *The variance conjecture on some polytopes*, arXiv:1209.4270 [1]. On PDF page 2, Conjecture 1.4 asks whether there is a universal C such that every centered log-concave random vector X satisfies

$$\mathrm{Var}(|X|^2) \leq C \lambda_X^2 \mathbb{E}|X|^2, \quad \lambda_X^2 = \|\mathrm{Cov}(X)\|_{\mathrm{op}}. \quad (1)$$

Conjecture 1.5 asks equivalently whether

$$\sigma_X := \left[\mathbb{E} \left(|X| - \sqrt{\mathbb{E}|X|^2} \right)^2 \right]^{1/2} \leq C \lambda_X. \quad (2)$$

The authors stress that these statements are not invariant under arbitrary linear maps and were not known to follow from the isotropic conjecture.

2 Later theorem used

Klartag and Lehec, *Thin-shell bounds via parallel coupling*, arXiv:2507.15495v2 [2], prove the following as Theorem 1.1 on PDF page 2: for every isotropic log-concave random vector Z in $\mathbb{R}^k$,

$$\mathrm{Var}(|Z|^2) = \mathbb{E}(|Z|^2 - k)^2 \leq C_0 k, \quad (3)$$

where C_0 is universal. Their statement ranges over every dimension k .

It is also known (see [BN], [EK]) that these two equivalent conjectures are stronger than the hyperplane conjecture, which states that every convex body of volume 1 has a hyperplane section of volume greater than some absolute constant.

The variance conjecture is a particular case of a stronger conjecture, due to Kannan, Lovász and Simonovits (see [KLS]) concerning the spectral gap of log-concave probabilities. This conjecture can be stated in the following way due to the work of Cheeger, Maz'ya and Ledoux:

Conjecture 1.3. *There exists an absolute constant C such that for any locally Lipschitz function, $g : \mathbb{R}^n \rightarrow \mathbb{R}$, and any centered log-concave random vector X in $\mathbb{R}^n$*

$$\text{Var } g(X) \leq C \lambda_X^2 \mathbb{E} |\nabla g(X)|^2.$$

Note that Conjecture 1.1 is the particular case of Conjecture 1.3 when we consider only isotropic vectors and $g(X) = |X|^2$. Our purpose in this paper is to study the particular case of Conjecture 1.3 in which $g(X) = |X|^2$ but X is not necessarily isotropic. Thus, we will study the following *general variance* conjecture:

Conjecture 1.4. *There exists an absolute constant C such that for every centered log-concave vector X*

$$\text{Var } |X|^2 \leq C \lambda_X^2 \mathbb{E} |X|^2.$$

In the same way that Conjecture 1.1 is equivalent to Conjecture 1.2, Conjecture 1.4 can be shown (see Section 2) to be equivalent to the following *general thin shell width* conjecture:

Conjecture 1.5. *There exists an absolute constant C such that for every centered log-concave vector X*

$$\sigma_X \leq C \lambda_X$$

Notice that since Conjecture 1.4 and Conjecture 1.5 are not invariant under lin-

Figure 1: Conjectures 1.4 and 1.5 in arXiv:1209.4270, PDF page 2.

3 Proof intuition

Whitening writes the squared norm of an arbitrary centered vector as a positive quadratic form $\langle MY, Y \rangle$ of an isotropic vector. Order the eigenvalues of M . Abel summation decomposes that quadratic form into a positive linear combination of the centered energies of nested orthogonal projections of Y . Each projected vector is again isotropic and log-concave, so the solved isotropic theorem controls each energy in its own dimension. Minkowski combines these controls without needing any sign information about mixed fourth moments. A one-line Cauchy–Schwarz estimate for the spectral counting function then produces exactly $\|M\|_{\text{op}} \text{tr } M$.

The positivity of the spectral increments is essential: it is what permits a constant-preserving passage from unweighted projection energies to every positive quadratic form.

For a random vector X in $\mathbb{R}^n$ with law μ we denote $\text{Cov}(X) = \text{Cov}(\mu)$. We say that X is log-concave (respectively, isotropic) if its law μ is log-concave (respectively, isotropic). It is well-known that for any random vector X with finite second moments whose support affinely spans $\mathbb{R}^n$, there exists an affine map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $T(X)$ is isotropic (see, e.g. [11] Section 2.3). Our main result is the following:

Theorem 1.1. *Let X be an isotropic, log-concave random vector in $\mathbb{R}^n$. Then,*

$$\text{Var}(|X|^2) = \mathbb{E}(|X|^2 - n)^2 \leq Cn, \quad (1)$$

where $C > 0$ is a universal constant.

Theorem 1.1 is tight, up to the value of the universal constant. Indeed, if X is a standard Gaussian random vector in $\mathbb{R}^n$ or if X is distributed uniformly in the cube $[-\sqrt{3}, \sqrt{3}]^n \subseteq \mathbb{R}^n$, then X is isotropic and log-concave with

$$\text{Var}(|X|^2) = Cn, \quad (2)$$

Figure 2: Klartag–Lehec, Theorem 1.1, arXiv:2507.15495v2, PDF page 2.

4 Constant-preserving transfer theorem

For clarity, define the optimal all-dimensional constants

$$C_{\text{iso}} = \sup_{k \geq 1} \sup_Z \frac{\text{Var}(|Z|^2)}{k},$$

$$C_{\text{gen}} = \sup_{n \geq 1} \sup_X \frac{\text{Var}(|X|^2)}{\|\text{Cov}(X)\|_{\text{op}} \mathbb{E}|X|^2},$$

where Z ranges over isotropic log-concave vectors and X ranges over nonzero centered log-concave vectors. If a law is supported on a proper linear subspace, all quantities are computed in its linear span.

Theorem 1 (Exact equivalence). *One has*

$$C_{\text{gen}} = C_{\text{iso}}.$$

Consequently, the Klartag–Lehec theorem proves (1) in every dimension, with no loss in the universal constant.

Proof. The inequality $C_{\text{iso}} \leq C_{\text{gen}}$ is immediate: for an isotropic vector in $\mathbb{R}^k$, the denominator in the definition of C_{gen} equals $1 \cdot k$.

We prove the reverse inequality. Fix a centered log-concave X and reduce to the linear span of its support. Put $M = \text{Cov}(X)$, which is then positive definite, and set $Y = M^{-1/2}X$. Thus Y is isotropic and log-concave, and

$$|X|^2 = \langle MY, Y \rangle, \quad \mathbb{E}|X|^2 = \text{tr } M.$$

Choose an orthonormal eigenbasis of M and write

$$\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n > 0, \quad \mu_{n+1} = 0.$$

Let P_k be the orthogonal projection onto the span of the first k eigenvectors. The linear image $P_k Y$ is log-concave, and its covariance on that k -dimensional subspace is the identity. Therefore $P_k Y$ is isotropic and, by the definition of C_{iso} ,

$$\left\| |P_k Y|^2 - k \right\|_{L^2} \leq \sqrt{C_{\text{iso}} k}. \quad (4)$$

If Y_i are the coordinates of Y in the eigenbasis, Abel summation gives the exact identity

$$\begin{aligned} \langle MY, Y \rangle - \text{tr } M &= \sum_{i=1}^n \mu_i (Y_i^2 - 1) \\ &= \sum_{k=1}^n (\mu_k - \mu_{k+1}) \sum_{i=1}^k (Y_i^2 - 1) \\ &= \sum_{k=1}^n (\mu_k - \mu_{k+1}) (|P_k Y|^2 - k). \end{aligned} \quad (5)$$

All coefficients in the final sum are nonnegative. Minkowski's inequality and (4) yield

$$\sqrt{\text{Var}(\langle MY, Y \rangle)} \leq \sqrt{C_{\text{iso}}} \sum_{k=1}^n (\mu_k - \mu_{k+1}) \sqrt{k}. \quad (6)$$

It remains to estimate a deterministic spectral sum. Let

$$N(t) = \#\{i : \mu_i \geq t\}, \quad 0 \leq t \leq \mu_1.$$

The layer-cake formula and Cauchy–Schwarz give

$$\sum_{k=1}^n (\mu_k - \mu_{k+1}) \sqrt{k} = \int_0^{\mu_1} \sqrt{N(t)} dt \quad (7)$$

$$\leq \left(\mu_1 \int_0^{\mu_1} N(t) dt \right)^{1/2} = \left(\mu_1 \sum_{i=1}^n \mu_i \right)^{1/2}. \quad (8)$$

Combining (6) with (7)–(8) and squaring proves

$$\text{Var}(|X|^2) \leq C_{\text{iso}} \mu_1 \sum_{i=1}^n \mu_i = C_{\text{iso}} \|M\|_{\text{op}} \text{tr } M = C_{\text{iso}} \lambda_X^2 \mathbb{E}|X|^2.$$

Hence $C_{\text{gen}} \leq C_{\text{iso}}$, completing the proof. $\square$

Corollary 2 (Source Conjectures 1.4 and 1.5). *There is a universal constant C such that every centered log-concave random vector satisfies both (1) and (2).*

Proof. The variance inequality follows from Theorem 1 and Klartag–Lehec's bound (3). For the thin-shell form, put $m = \mathbb{E}|X|^2$. Pointwise,

$$(|X| - \sqrt{m})^2 \leq \frac{(|X|^2 - m)^2}{m},$$

because $|X| + \sqrt{m} \geq \sqrt{m}$. Consequently,

$$\sigma_X^2 \leq \frac{\text{Var}(|X|^2)}{\mathbb{E}|X|^2} \leq C \lambda_X^2.$$

Taking square roots proves Conjecture 1.5 after renaming the universal constant. $\square$

5 Verification and scope

The proof uses only four points: whitening; closure of log-concavity under linear images; the Klartag–Lehec theorem in each projected dimension; and the deterministic Abel identity and spectral estimate displayed above. The accompanying seeded script checks the Abel identity and spectral inequality on 20,000 random nonnegative spectra. This computation is an algebra smoke test, not evidence for the Klartag–Lehec theorem.

The conclusion settles the source’s Conjectures 1.4 and 1.5. It does *not* prove the stronger Kannan–Lovasz–Simonovits conjecture (source Conjecture 1.3), which concerns every locally Lipschitz test function rather than only $g(x) = |x|^2$.

6 Literature and novelty status

This is classified as a literature-implied answer because the only deep input is the later theorem [2]. The supporting authors do not state that they answer arXiv:1209.4270, and the constant-preserving spectral transfer is the agent-identified implication.

A bounded search on 22 July 2026 covered the run indexes; the full TeX of [2]; exact arXiv/web searches for “general variance conjecture”, the displayed anisotropic inequality, and quadratic-form versions; and the later Alonso-Gutierrez–Bastero paper arXiv:1610.04023 [3]. The latter still says that equivalence between the isotropic and general conjectures was unclear and proves special families. No later paper explicitly stating Theorem 1 or the full anisotropic conclusion was found. Because the transfer lemma is short, it may nevertheless be folklore; novelty confidence is moderate.

7 Human review recommendation

Treat this as a full literature-implied answer. Reviewers should verify the constant-preserving Abel decomposition, the use of the isotropic theorem on all orthogonal marginals, and the provenance/novelty classification. The mathematical scope is all centered log-concave laws and includes degenerate laws after restriction to their linear span.

References

- [1] D. Alonso-Gutierrez and J. Bastero, *The variance conjecture on some polytopes*, arXiv:1209.4270 (2012; PDF revision dated 2018).
- [2] B. Klartag and J. Lehec, *Thin-shell bounds via parallel coupling*, arXiv:2507.15495v2 (2026), Theorem 1.1.
- [3] D. Alonso-Gutierrez and J. Bastero, *The variance conjecture on hyperplane projections of the ℓ_p^n balls*, arXiv:1610.04023; Rev. Mat. Iberoam. 34 (2018), 879–904.